

Summary

Final-year B.Tech ECE student with a strong foundation in **VLSI Design**, **Embedded Systems**, and **Quantum Engineering**. Hands-on experience in circuit design (advanced technology nodes), firmware development (RISC-V architecture), and research (metasurfaces & quantum correlations). Eager to leverage strong technical background and problem-solving skills in high-growth engineering teams.

Education

PES University, Bengaluru

Expected Graduation: 2026

B.Tech in Electronics and Communication Engineering (Major) — Minor in Computer Science

Relevant Coursework: Robotics, VLSI Design, Quantum Mechanics, Embedded Systems, Digital Signal Processing

Technical Skills

- **VLSI & Chip Design:** Cadence Virtuoso, Cadence Suite, Xilinx Vivado, Digital Logic Design
- **Hardware & Embedded:** RISC-V Architecture, Microcontroller Programming, PCB Design
- **Simulation & Design Tools:** Lumerical, Tanner Tools, RISE, Wireshark, VS Code, Git
- **Programming:** Verilog, SystemVerilog, Assembly, C, C++, Python, JavaScript, SQL
- **Systems:** Communication Protocols, Real-time Systems, Linux, Static Timing Analysis

Experience and Projects

Research Intern in L&T funded project

- Design of the electrical architecture for an industrial robotic system, including power distribution, motor drivers, and sensor fusion.
- NVIDIA Jetson Nano Deployment for embedded processing and hardware-software integration on an automated industrial machine.
- Implemented sensor interface (LiDAR, IMU, encoders) and closed-loop motor control for precision industrial operation.

Advanced Circuit Design and Simulation

2023–2024

- Created & simulated custom digital logic using modern VLSI methodologies.
- Used Cadence Virtuoso & Tanner Tools for schematic/layout and functional verification.
- Focused on power/performance optimization in semiconductor design flows.

RISC-V Based Embedded System Development

2023

- Developed firmware for RISC-V with real-time data handling & signal processing.
- Wrote optimized C/Assembly for high-performance I/O; implemented interrupt-driven logic.
- Built robust system-level control with attention to reliability and latency.

Cadence® Tempus™ Technology – Basic Static Timing Analysis v3.0 certified.

2025

- Demonstrated understanding of timing concepts like setup and hold checks, clock uncertainty, and multicycle paths using Cadence Tempus™, a leading EDA tool for timing verification in digital design.
- Certified in applying On-Chip Variation (OCV) and constraint-driven analysis to ensure accurate and reliable timing across process, voltage, and temperature (PVT) corners.

Core ECE Subjects

Strongest Subjects

- VLSI Design
- Quantum Mechanics
- Embedded Systems
- Digital Signal Processing

Core Concepts

- Communication Theory
- Circuit Analysis
- Digital Logic Design
- Control Systems

Core Competencies & Soft Skills

Technical

- Problem Solving
- Analytical Thinking
- System Design & Verification
- Hardware-Software Integration
- Research & Verification Development

Soft Skills

- Team Collaboration
- Communication
- Innovation & Leadership
- Creative Problem-Solving
- Time Management

Achievements & Certifications

- **Cadence® Tempus™ Technology – Basic Static Timing Analysis v3.0** certified (setup/hold, uncertainty, multicycle paths, OCV).
- Achieved **Artificial Intelligence & Machine Learning** certification.
- Award-winning robotics performance through systematic embedded systems design.
- Multiple hackathons focused on hardware-software integration and rapid prototyping.
- Game development hackathons using **Unreal Engine** and **GODOT**.
- Experience with financial analysis and market research via stock market competitions.